

COBHAM-FREE FULL GENERAL AUTOMATIC RIGIDITY FOR COLLATZ-INVARIANT SEQUENCES

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Abstract. Let the shortcut Collatz map on nonnegative integers be defined by $T(2n)=n$ and $T(2n+1)=3n+2$. Earlier ARC developments established positive-domain rigidity for several individual bases, all odd bases, and all even bases, but the integrated all-base Lean theorem still retained an explicit Cobham-principle parameter through the base-2 / pure-power-of-two branch. This note removes that final dependence.

The key new step is a Cobham-free reconstruction of ARC2. For a base-2 finite-kernel automatic coloring a satisfying $a(T(n))=a(n)$, define the odd subsequence $b(n)=a(2n+1)$. The original ARC2 kernel arithmetic gives base-3 finite-kernel automaticity of b . A local-density theorem for shortcut-Collatz invariance yields dyadic nowhere-thickness of the adjacent-change set of b . The general odd-base topological-pumping theorem at base 3 then forces that change set to be finite, hence b is eventually constant. The identity $b(4n+2)=b(n)$ propagates the tail value to every index of b , and doubling invariance propagates the result to every positive index of a .

Combining this new ARC2 endpoint with power-of-two factor descent and the previously verified odd-base General ARC gives: for every base $B \geq 2$, every base- B finite-kernel automatic shortcut-Collatz-invariant coloring is constant on the positive integers. The resulting Lean endpoint has no `ARCCobhamPrinciple` parameter, no `[Finite alpha]` typeclass assumption, and the final audit reports only the standard Lean/Mathlib dependencies `propext`, `Classical.choice`, and `Quot.sound`; no `sorry` occurs. A full project build completed successfully with 8906 jobs.

1. Purpose and relation to earlier ARC releases

This note is a new release in the ARC series. It does not replace the earlier ARC2, ARC3, ARC5, ARC7, ARC15, all-even-base, or general odd-base papers. Those papers are retained as separate stages of the development. The present result is not a correction of a false theorem; rather, it removes an external hypothesis that remained in the earlier integrated proof architecture and thereby strengthens the final all-base theorem.

The central rigidity problem is: let a be a coloring of the nonnegative integers and suppose it is invariant under the shortcut Collatz map. Equivalently:

$$a(2n) = a(n), \quad a(2n+1) = a(3n+2).$$

For a base $B \geq 2$, define the B -kernel $K_B(a)$ as the family of sequences $n \rightarrow a(B^e n + r)$ with $e \geq 0$ and $0 \leq r < B^e$. We say that a is base- B finite-kernel automatic when this family is finite.

Theorem 1.1 (Full General ARC, Cobham-free). Let $B \geq 2$ and let a be a coloring of the nonnegative integers. Assume $K_B(a)$ is finite and $a(T(n))=a(n)$ for every $n \geq 0$. Then there exists a value c such that $a(n)=c$ for every $n > 0$. No conclusion is forced at $n=0$.

The point of this note is that Theorem 1.1 is obtained in Lean without an explicit appeal to Cobham's theorem. Cobham's theorem remains fundamental in automatic-sequence theory, but it is no longer a dependency of the present final ARC proof.

2. The old bottleneck: the pure power-of-two branch

The general odd-base development already gave positive-domain rigidity for every odd base $m \geq 3$. For an even base $B=2^s m$ with $s > 0$ and m odd, doubling invariance gives a factor-descent theorem:

$$(2^s m)\text{-finite-kernel automatic} \implies m\text{-finite-kernel automatic}.$$

Hence every mixed even base with $m > 1$ already reduced to the odd-base theorem without Cobham. After localization, the only remaining branch was $m=1$, namely $B=2^s$. Factor descent reduces that branch to base 2. Thus the entire all-base theorem becomes Cobham-free once ARC2 itself is reconstructed without Cobham.

3. Cobham-free ARC2

Let a be base-2 finite-kernel automatic and shortcut-Collatz invariant. Define $b(n)=a(2n+1)$. The proof separates into four ingredients.

3.1 Base-3 automaticity of the odd subsequence

The original ARC2 arithmetic shows that the base-3 kernel of b embeds into the finite base-2 kernel generated by a . In the formal development this is exposed by `arc_three_kernel_odd_finite`, and the new wrapper is `arc2OddSubseq_threeAutomatic_of_twoAutomatic`. This lower-level route does not require a `[Finite alpha]` instance.

3.2 Dyadic thinness of the change set

Define $D_b = \{n : b(n) \neq b(n+1)\}$. The new insight is that no Cobham-style periodicity theorem is needed. The local-density theorem for invariant colorings gives, inside every dyadic parent cylinder, a deeper child on which $a(x)=a(x+2)$. Under the odd affine image $x=2n+1$, this is exactly $b(n)=b(n+1)$. Therefore every dyadic parent contains a deeper child disjoint from D_b ; the change set is dyadically nowhere thick. This is formalized by `arc2OddSubseq_changeSet_nowhereThick` and packaged as `arc2CobhamRemoval_stage1_thinness`.

3.3 Finite change set and eventual constancy

The General Odd ARC machinery contains the required topological-pumping theorem at base 3: a base-3 finite-kernel automatic change indicator with dyadically nowhere-thick support has finite support. Consequently D_b is finite, so b is eventually constant.

3.4 Propagation from the tail

Shortcut invariance gives $b(4n+2)=b(n)$. Iterating $n \rightarrow 4n+2$ sends every starting index to arbitrarily large indices while preserving b . Hence eventual constancy implies global constancy of b . Finally, every positive integer can be reduced to an odd integer by repeatedly using $a(2n)=a(n)$. This proves the Cobham-free ARC2 endpoint `arc2_main_kernel_form_no_cobham`.

4. Closing every base

Proposition 4.1 (Pure powers of two). Let $B=2^s$ with $s>0$. If a is base- B finite-kernel automatic and shortcut-Collatz invariant, factor descent gives base-2 automaticity; the Cobham-free ARC2 theorem then gives positive-domain constancy.

Proposition 4.2 (Mixed even bases). Let $B=2^s m$ with $s>0$, m odd, and $m>1$. Factor descent gives base- m finite-kernel automaticity, and the General Odd ARC theorem applies directly because $m \geq 3$ is odd.

Proof of Theorem 1.1. If B is odd, then $B \geq 3$ and General Odd ARC applies. If B is even, factor $B=2^s m$ with $s>0$ and m odd. Use Proposition 4.1 when $m=1$ and Proposition 4.2 when $m>1$. These cases exhaust all $B \geq 2$. The packaged Lean endpoint is `arc_full_general_no_cobham`.

5. Lean 4 formalization and final audit

Active Cobham-removal chain.

1. `ARC2CobhamRemovalStage0`. lean - odd-subsequence packaging, tail-to-global propagation, and positive-domain recovery. Endpoint: `arc2OddSubseq_rigidity_of_threeAutomatic_and_thin`.
2. `ARC2CobhamRemovalStage1_Thinness_v3`. lean - dyadic nowhere-thickness from local density. Endpoint: `arc2CobhamRemoval_stage1_thinness`.
3. `ARC2CobhamRemovalStage2_Final_v2`. lean - lower ARC2 kernel transfer and Cobham-free ARC2 closure. Endpoint: `arc2CobhamRemoval_stage2_final`.
4. `ARC2CobhamRemovalStage3_FullGeneral`. lean - replacement of the pure power-of-two branch and all-base closure. Endpoint: `arc_full_general_no_cobham`.
5. `ARCFullGeneralNoCobhamFinalAudit`. lean - independent all-base reconstruction and representative smoke tests. Endpoint: `arcFinalAudit_full_general_no_cobham`.

The final audit includes concrete instances at bases 2, 3, 4, 5, 6, 7, 10, 15, 64, and 100. For the principal final theorems, `#print axioms` reports `propext`, `Classical.choice`, and `Quot.sound`, and does not report `sorryAx`. The project-wide command `lake build` completed successfully with 8906 jobs after the audit file was imported into the root project.

Audit scope. The axiom report is a formal dependency audit, not an independent mathematical peer review. It verifies the encoded proof chain relative to the imported Lean/Mathlib environment and definitions.

6. What changed, and what did not

The earlier ARC papers remain useful and are intentionally retained. ARC2 recorded the original base-2 route and kernel arithmetic. ARC5, ARC7, ARC15, and General Odd ARC developed the local-density, dyadic-thinness, pumping, finite-change, and positive-rigidity machinery that ultimately made the new ARC2 proof possible. The all-even and first Full General integrations exposed exactly where the external Cobham handoff remained.

The present note therefore records a proof-architecture improvement: machinery developed later in the ARC program is fed back into the earliest base-2 case, eliminating the last explicit Cobham dependency from the integrated all-base theorem. This does not imply that Cobham's theorem is unnecessary in automatic-sequence theory generally, nor does it prove the Collatz conjecture.

7. Sharpness at zero

The positive-domain conclusion is sharp. Given arbitrary values x and y , define $a(0)=x$ and $a(n)=y$ for $n>0$. This coloring is finite-kernel automatic in every base and is shortcut-Collatz invariant, while $a(0)$ is independent of the common positive value. Thus Theorem 1.1 cannot in general be strengthened to include $n=0$.

8. Release note

First released: September 23, 2026. Earlier ARC papers are not superseded or withdrawn. They are retained as historical stages of the development. This paper is added as a new release documenting the Cobham-free closure of the Full General ARC theorem.

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References. [1] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003. [2] A. Cobham, On the base-dependence of sets of numbers recognizable by finite automata, *Math. Systems Theory* 3 (1969), 186-192.

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